JC2 H1 Prelim 2024
Suggested Answers

Section A Cluster 2 Fieldwork

1. A group of 10 students wanted to investigate if businesses within the community is actively responding to climate change.

They visited Tampines IKEA (a mega home furnishing store) to find out their green initiatives, and did an investigation on people's awareness and response to climate change actions in Tampines IKEA over a Saturday afternoon from 3pm-5pm where the shop is expected to be crowded to get a bigger sample size.

They developed a research question: ***Are businesses in Singapore doing enough to encourage climate change action?***

The students went around Tampines IKEA in groups of 2 to observe the store and approached any person who was willing to do their questionnaire survey. They surveyed 20 persons in all, including one staff member. Most of those they surveyed were young couples who were there to get furniture for their new homes. They also went round to the showrooms and the restaurant to take photographs of areas they feel the business was doing enough to mitigate climate change.

Resource 1 shows pictures the students took of areas in Tampines IKEA where they felt the business was doing enough to mitigate climate change. Resource 2 shows the questionnaire that the students used to survey some of the shoppers there. Resource 3A and 3B shows the results of the questionnaire survey that the students did to show their questionnaire findings.

(a) Explain two weakness of the student's research question. [4]

Award 1 mark for each explanation of the weakness of student's research question, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Student's research question is too broad/ not specific when they use "*doing enough*" as there is no clear measurement of what enough is and what is not [1]. Moreover, the type of business is not specified. [1 additional mark]

- Students' research is unclear as "climate change actions" has many types and at many different levels. [1]

- Students' research is too large a scale as "businesses in Singapore" can refer to many business/ types of businesses [1] that is too much for 10 students to do an investigation on. [1 additional mark]

(b) With reference to Resource 1, explain two possible impacts the students might have while doing the investigation. [4]

Award 1 mark for each explanation of the possible impacts, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each impact, where applicable.

Possible responses include:

- Impact on **business** – the students will impact the smooth running of the business if they stop shoppers to do their interview [1]. This is especially so if they survey staff members or station themselves at inappropriate places like near the shuttle bus area (R1) or near checkout points and interfere with the flow of human traffic. [1 additional mark]

- Impact on **shoppers and diners** at IKEA – the students might be interfering into the privacy or personal time of the shoppers who might be wanting a peaceful meal by themselves in IKEA (R1) [1]. This is especially so if students wander into the eating places to do their surveys or approach couples who are seated down to do planning for their homes. [1 additional mark]

- Positive Impact on **shoppers** – raise awareness might be curious of what the students are researching about and might ask [1]. When the students explain, those who do not know of climate initiatives will get to know and be aware. [1 additional mark]

(c) Suggest how the students could minimise <u>three</u> potential risks while doing their investigation at Tampines IKEA. [6]

Award 1 mark for each suggestion of a potential risk, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each risk, where applicable.

Possible responses include:

- Risk from **crowd** - because there are many people there on a weekend, and pushing large furniture items around, the students might be knocked into when they stop to survey a shopper [1]. Instead, they could ask the person they are surveying to maybe move to a safer place where there is not much human traffic. [1 additional mark]

- Risk from **falling objects/ sharp objects** – because a lot of the furniture is stacked (R1) and there may be many people, the students might knock over these and injure themselves if they are not aware of the displays, especially when their focus is on looking for people to survey ***OR*** they might cut themselves near the recycling station with broken pieces of glass or plastic not properly disposed [1]. Instead, they should stay clear from areas that are crowded and station themselves maybe outside IKEA to do the survey instead ***OR*** don't touch the recycled are unnecessarily. [1 additional mark]

- Risk of **accidents** – especially when the students go outside of IKEA to take photographs of the surroundings (eg. the bus area) as there will be vehicular movements and also shoppers transporting their bulky items onto their vehicles [1]. They should look out for traffic when on the roads and not go near shoppers carrying bulky items, just in case they get hit. [1 additional mark]

- Risk of **angry shoppers**– as some shoppers who are unsure of the legitimacy of the students doing a survey or might not want to be interrupted while shopping, the students run the risk of an angry shopper shouting at them or being unreasonable [1]. Thus students should approach IKEA first to get permission to do the survey in store – preferably with some identification tag to proof their legitimacy. [1 additional mark]

- Risk of **getting lost** due to the big area and its crowded [1] – ensure remain contactable via HP or meeting point if lost each other. [1 additional mark]

(d) With reference to Resource 3A, explain how the students can improve the data presentation method. [6]

Award 1 mark for each explanation of how students can improve the data presented, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each improvement, where applicable.

Possible responses include:

- As the table in R3A does not give a visual presentation of the different responses [1], the students might want to **present their data in a bar chart** instead, so that the responses can be **compared easily** – especially for Q4 that reflects the mode of transport with more than 2 options. [1 additional mark]

- Number of respondents for each question can be indicated over each of the bars in the bar chart to allow reader to **easily see/ visualize** the actual number [1]

- Bar graph of 4 questions can be drawn together with y-axis all being “the number of respondents” so that it is **easy to compare** across questions. [1]

- It is also not useful to put the responses of the 3 questions together (R3A) as they are not related and it makes the data presentation confusing [1]. The students should instead **present the 3 sets of data separately** so that it is **clear** that they relate to different aspects of shopper’s response. [1 additional mark]

- Present age demographics also to show range of age groups represented [1].

(e) Based on observations in Tampines IKEA and the results from the questionnaire survey, the students concluded that businesses in Singapore are doing enough to encourage climate change action. Evaluate the validity of their conclusion. [10]

Possible Approaches

The students' conclusion is ***VALID*** because:

- IKEA seems to be doing a fair bit to encourage climate change action from the collection of photographs the students have in **R1**. There are recycling stations for shoppers to drop off their recyclables, recyclable shopping bags for shopper's use and the plant based meals and use of public transport also encourage lower carbon foot-print.

- The result of the student's survey in **R3A** also shows that 16 out of the 20 respondents to the survey feel that IKEA is environmentally friendly, this majority of the responses show that the student's conclusion is valid. The respondents are also aware of the green initiatives in Tampines IKEA, thus the business can be said to encourage climate change action.

However, there are some ***limitations*** of the investigation that diminishes the validity of the student's conclusion:

- The **research question** was on businesses in Singapore, yet the students only managed to investigate one business, IKEA. The number is too small to represent businesses as a whole in Singapore, thus the conclusion is not fully valid. The students should spend more time and investigate more businesses instead, and businesses of various sectors – eg. Electronics, F&B, etc. otherwise, they can change their research question to just IKEA, then their conclusion will be more valid.

- **R3B** also reveals that some respondents come to IKEA for the air-conditioning, which is not exactly promoting climate change action. For some this can be seen as greenwashing by businesses. However, the comfortable environment does increase awareness of IKEA's sustainable products and food, thus the conclusion has some validity.

- The survey the students used (**R2**) is also short with only 4 questions. It does not collect enough information to show if businesses in Singapore is doing enough to encourage climate change. The student should instead look for reliable secondary sources that has official data on what business (like IKEA) has done to encourage climate change action.

- The survey the students used in **R2** just sampled anyone who was willing to do it (convenience sampling), so could be biased as young couples could seem most approachable compared to some elderly, thus sample is bias and thus the conclusion is not reliable and not fully valid.

Section B Cluster 1 Sustainable Future and Climate Change

2. Resource 4 shows the proportion of global urban population living in slums (percentage) and total slum population (millions) between 2000 and 2020. Resource 5 is a photograph of Happyland, part of the Tondo slum in Manila. Resource 6 shows the relationship between GDP per capita and the prevalence of undernourishment and TB incidence per 100 000 population in 2021.

(a) Describe the changes in the global urban population living in slums between 2000 and 2020 as shown in Resource 4. [4]

Award 1 mark for each description of the changes in the global slum population, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The proportion of global urban population living in slums has **decreased** from approximately 31% in 2000 to about 24% in 2020. [1]
- The change in global urban population living in slums has been a **steady decline**, although it starts to **stabilise from 2016 to 2020**. [1 additional mark]
- On the other hand, the total number of people living in slums has **increased significantly** from close to 900 million people in 2000 to approximately 1,060 million in 2020. [1]
- Within the first five years from 2000 to 2005, the change has been a **steady increase** of 50 million. However, the gradient **flattens between 2005 to 2016**, suggesting a slowdown in the growth of slum population, before it **rises steeper again from 2016 to 2020 suggested a renewed increase**. [1 additional mark]

(b) Suggest reasons to explain the trends in global slum population in Resource 4. [5]

Award 1 mark for each suggested reason to explain the trends in global slum population, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

- The proportion and total numbers can show opposite trends because they are **influenced by different rates of growth**. [1] The proportion is a **relative** measure, while the total number is an **absolute** measure. [1 additional mark]

- Even if the percentage of people living in slums decreases over time, the **global urban population might be growing rapidly** enough that the total number of people living in slums still rises. [1] This is **due to factors** such as rural-to-urban migration, natural population growth in cities, and urbanisation. [1 additional mark]

- **Rapid urbanisation** in developing countries often **outpaces the development of adequate housing and infrastructure**. [1] As more people move to cities seeking

better opportunities, they may end up living in slums due to the **lack of affordable housing**, leading to an increase in the total number of people living in slums. [1 additional mark]

- Economic growth in urban areas can lead to an **increase in income inequality**. [1] While some urban residents might move out of slums due to rising incomes, the **influx of poorer migrants** or the **population growth among those already in slums** might continue to add to the slum population. [1 additional mark]

(c) With reference to Resource 5, describe three aspects of the deprivation experienced by slum dwellers in Tondo, Manila. [3]

Award 1 mark for each description of the deprivation experienced by slum dwellers in Tondo, to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The ground is littered with debris and waste, suggesting **poor sanitation**. [1]
- The dwellings appear to be **makeshift structures** made from a mix of materials like corrugated metal, wood, and tarpaulins. [1]
- The narrow, congested alleyway filled with people indicates **overcrowding**. [1]
- The visible power lines strung haphazardly across the area indicate a potential **lack of formal electrical connections**, which can be dangerous and unreliable. [1]

(d) Describe the relationship between GDP per capita and the prevalence of undernourishment and TB incidence as shown in Resource 6. [4]

Award 1 mark for each description of the relationship between GDP per capita and the prevalence of undernourishment and TB incidence, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The two scatter plots in Resource 6 describe the relationships between GDP per capita, the prevalence of undernourishment, and TB incidence.

- There is a **clear negative correlation between GDP per capita and TB incidence**. [1] As GDP per capita increases from US$1000 to US$100,000, the incidence of TB (measured per 100,000 population on a log scale) decreases from 700 to 7. [1 additional mark]

- There is a **positive correlation between the prevalence of undernourishment and TB incidence**. [1] As the prevalence of undernourishment (measured as a percentage of the population) increases from 3% to 30%, the incidence of TB also increases from 20 to 300. [1 additional mark]

(e) Explain how tuberculosis (TB) can spread in slums, such as that shown in Resource 5. [6]

Award 1 mark for each explanation of how tuberculosis (TB) can spread in slums, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Slums are often characterised by **high population density and overcrowded living conditions**, where many people live in **close proximity** within **small spaces**. [1] TB is an **airborne disease**, and overcrowding facilitates the spread of TB bacteria through droplets when an infected person coughs, sneezes, or even talks. [1 additional mark]

- Many slum dwellings have **inadequate ventilation**, which allows the **TB bacteria to linger in the air for longer periods**. [1] **Poor air circulation** means that the bacteria are more likely to be inhaled by others, increasing the risk of transmission, especially in crowded indoor environments. [1 additional mark]

- Slum dwellers often experience **malnutrition, inadequate access to healthcare, and high levels of stress due to poverty**. [1] These factors can **weaken the immune system**, making individuals more susceptible to contracting TB and less capable of fighting off the infection once exposed. [1 additional mark]

- In slums, **access to healthcare services is often limited** due to financial constraints, lack of nearby facilities, or overcrowded health services. [1] This can lead to **delays in diagnosing and treating TB**, allowing the disease to progress and spread to others. Additionally, people with undiagnosed or untreated TB can **unknowingly transmit** the bacteria **to others** in the community. [1 additional mark]

3. Resource 7 shows the number of days with temperature increase above pre-industrial levels between 1990 to 2023. Resource 8 shows climate change impact on various aspects of nature. Resource 9 shows the global climate change vulnerability index in 2017.

(a) Describe the changes in temperature from 1990 to 2023 as shown in Resource 7. [4]

Award 1 mark for each description of the changes in temperature, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Before 1995, there were some years without days where temperatures increase above pre-industrial levels of 1-1.5◦, like the years 1992-94 [1].

- After 1995 to 2014, most years have days where temperatures increase above pre-industrial levels of 1-1.25◦, with some years where temperatures increase above pre-industrial levels of 1.25-1.5◦ [1], although most of these years have less than 200 days that was within this range. [1 additional mark]

- However, from 2015 onwards till 2023, all years had more than 200 days where temperatures increase above pre-industrial levels of 1-1.25◦, 1.25-1.5◦ and even more than 1.5◦[1], with 2023 having by far the most number of days (almost half the year) where temperatures was above 1.5◦ or more. [1 additional mark]

- For 2019 and 2023, all the days of the year (365 days) had temperature increase above pre-industrial levels of 1-1.25◦, 1.25-1.5◦ and even more than 1.5◦. [1]

(b) With reference to Resource 8, describe how climate change can impact nature in two areas. [4]

Award 1 mark for each description of how climate change can impact nature, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Climate change leading to global warming can impact nature by reducing snow cover in the spring and summer [1]. This loss in snow cover increase melting and evaporation, and can sometimes lead to increased floods. [1 additional mark]
- Climate change can also lead to decreased river flow, like that in southern California [1]. Decreased river flow can affect the biodiversity of rivers – especially if the reduced flow leads to a lack of oxygen in the water that can cause fishes to die or even become extinct. [1 additional mark]
- Climate change can also lead to ocean acidification when too much CO2 is absorbed into the water, making it hard for some species to build shells [1].

(c) Explain how the two areas identified in part (b) can impact the human system. [4]

Award 1 mark for each explanation of how each area can impact the human system, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- When there is snow cover loss due to melting and evaporation, the water supply of a region can be affected [1]. This is because increased evaporation (due to global warming) from the rivers as it flows downstream reduces the discharge [1]. **OR** For example, the Himalayas are the water towers of Asia and provides much needed water for the Asia region. Climate change has lowered the ability of the Himalayas to provide water for downstream countries. [1 additional mark]

- With increase warming, the increased risk of floods can affect populations living downstream when river floods destroy homes and communities [1]. This is because often, populations chose to locate next to river for their livelihood, and when river flood due to increase snowmelt, these communities are destroyed. [1 additional mark] **OR** For example, flash floods in Pakistan on Shounter Valley caused land erosion and extensive damage to crops and residential homes. [1 additional mark]

- A decreased river flow that leads to reduction in fish number and species can affect fishing communities who depend on it for food and livelihood [1]. This is because with reduced fish, there is reduced food supply and income for these communities, leading to economic hardship. [1 additional mark] **OR** An example will be how the indigenous peoples in America and Canada were affected when the salmon industry that supported thousands of jobs was impacted by a reduction in salmon population. [1 additional mark]

(d) Describe the pattern of climate change vulnerability across the world in 2017 as shown in Resource 9. [4]

Award 1 mark for each description of the pattern of climate change vulnerability, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Countries that were most vulnerable (at extreme risk) to climate change across the world in 2017 are usually within the **tropics**, like South Sudan, the Central African Republic, the Democratic Republic of the Congo and Haiti [1]. They are also countries that are **poor,** and for Haiti, a **SIDs** (small island developing states). [1 additional mark]

- Countries that were least vulnerable (at low risk) to climate change across the world in 2017 are usually the countries at **higher latitudes**, like Denmark, the UK, Iceland, Ireland and Uruguay [1]. They are also countries that are **richer**, and more developed. [1 additional mark]

(e) With reference to Resource 9, suggest why some countries are more vulnerable to climate change than others. [6]

Award 1 mark for each suggestion to why some countries are more vulnerable, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

- Countries like those in Africa (Liberia, Central African Republic, South Sudan and the Democratic Republic of Congo) are more vulnerable because they are **poor** countries [1]. This limits their ability to **adapt** to climate change that richer countries can – eg. air-conditioning, adequate shelter, climate resilient homes, etc. [1 additional mark]

- Countries like Haiti, a small island developing state, is more vulnerable to climate change because it is a **low lying** coastal plain, and is exposed to the Atlantic hurricane belt [1]. This makes it very prone to floods due to extreme weather conditions or **rising sea level** as climate changes. [1 additional mark]

- These poor countries that are most vulnerable are also very much **dependent** on their natural environment for livelihood and survival especially rain-fed agricultural communities [1]. Traditional precipitation patterns can be **disrupted** by climate change, resulting in increased frequency and intensity of extreme weather events like droughts, floods, and storms. These can cause water scarcity, soil erosion, nutrient leaching, and waterlogging, all of which negatively impact crop productivity and can lead to **crop failures,** affecting livelihood and economy. [1 additional mark]

Section C Cluster 1 Sustainable Future and Climate Change

4. 'Progress towards sustainable urban development is hampered by high urbanisation rates and rapid urban growth.' Evaluate this statement. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether progress towards sustainable urban development is hampered by high urbanisation rates and rapid urban growth or urban population loss. Candidates could consider the challenges posed by the urban population trends such as pollution, strain on urban services, slums, poverty, urban blight, underutilisation of infrastructure etc. Candidates could analyse how the challenges hampers progress towards sustainable urban development in terms of the **scale of the challenges**, the **complexity of the challenges**, **how the challenges may have evolved over time** and the **stakeholders impacted (gender, age income level)**.

Candidates could also approach the question by making a judgement on whether progress towards sustainable urban development is hampered by high urbanisation rates and rapid urban growth in some contexts than others through a consideration of two or more case studies. Candidates could analyse how progress towards sustainable urban development can also be hampered by urban population loss. Candidates could analyse the contextual factors such as **level of economic development**, **resources available, demographic structure, physical environment,** & **political will**, which may have contributed to the differences in the progress towards sustainable urban development.

Levels marked using Generic Level Descriptors for H1 essays

5. Evaluate the issues that affect the liveability of elderly living in the cities. [13]

Possible approaches:

Candidates could approach the question by making a judgment on the relative significance of the issues that affect the liveability of elderly living in the cities. Candidates could consider **issues related to economic well-being, social well-being and psychological well-being**. Candidates could analyse the issues in terms of the **spatial and temporal scale, cultural setting, and social structure** etc. Candidates could also analyse whether the different types of issues are within their own control or not.

Candidates could also approach the question by making a judgment on the different issues that affect the liveability of elderly in different contexts. This can be done through a consideration of **two or more case studies** where issues were different in cities due to local variations in relation to their **economic, social, physical and political background**. Candidates could analyse the contextual factors such as **level of economic development**, **resources available, demographic structure, cultural norms, physical environment,** & **political will**, which may have contributed to the differences in the relative significance of the issues.

Levels marked using Generic Level Descriptors for H1 essays

6. 'It is possible to implement effective strategies to respond to contemporary climate change.' Evaluate this statement. [13]

Possible approaches:

Candidates could approach the question by making a judgment on the possibility of implementing effective strategies to respond to contemporary climate change in some contexts than others through a consideration of **two or more case studies**. Candidates could evaluate what enabled the strategies in each case study to be implemented/ not implemented. Candidates could analyse the **contextual factors such as the geography of the contexts, the scale of the problem, the political will of the authorities and the resources available**, which may have contributed to the extent and possibility of implementing climate change management strategies in these case studies.

Candidates could also approach the question by making a judgement on the possibility of implementing effective **mitigation and adaptation strategies** through a consideration of **criteria such as spatial scale of the strategies, amount of carbon captured, costs of the strategies, ability to bring about long-term solution and ability to address root causes of climate change**. Candidates could also analyse the **challenges in implementing strategies** to respond to contemporary climate change.

Levels marked using Generic Level Descriptors for H1 essays